Java Interview Questions & Answers (Fresher Guide)

1. What is Java?

Java is an object-oriented, platform-independent programming language.

2. JDK vs JRE vs JVM

JDK contains tools for development, JRE runs Java apps, JVM executes bytecode.

3. Compilation Process

Java source -> bytecode (.class) via javac -> JVM executes.

4. Platform Independent?

Bytecode runs on any JVM regardless of OS.

5. public static void main

Entry point of a Java application.

6. Primitive vs Non-Primitive

Primitive store values directly; non-primitive store references.

7. Access Modifiers

public, protected, default, private.

8. == vs equals()

== compares references; equals() compares content.

9. Pass by Value?

Java is always pass-by-value.

10. Why String Immutable?

Security, thread safety, and string pool optimization.

11. Four OOP Pillars

Encapsulation, Inheritance, Polymorphism, Abstraction.

12. Abstraction vs Encapsulation

Abstraction hides implementation; encapsulation hides data.

13. Inheritance vs Composition

Inheritance is IS-A; Composition is HAS-A.

14. Overloading vs Overriding

Overloading: same method name, different parameters. Overriding: redefine inherited method.

15. Runtime Polymorphism

Method overriding resolved at runtime.

16. Abstract Class vs Interface

Abstract classes can have state; interfaces define contracts.

17. Interface Implementation

Yes, default and static methods can have implementations.

18. Multiple Inheritance

Supported through interfaces.

19. String Pool

Special memory area storing reusable string literals.

20. String vs StringBuilder vs StringBuffer

String immutable; Builder fast; Buffer synchronized.

21. Garbage Collection

Automatically removes unused objects.

22. Stack vs Heap

Stack stores method calls/local vars; Heap stores objects.

23. References

Strong survives GC; Weak/Soft may be reclaimed.

24. Checked vs Unchecked Exceptions

Checked handled at compile-time; unchecked at runtime.

25. throw vs throws

throw throws exception; throws declares it.

26. Multiple Catch

Yes, multiple catch blocks are allowed.

27. finally Block

Executes regardless of exception occurrence.

28. ArrayList vs LinkedList

ArrayList faster access; LinkedList faster insertion/deletion.

29. HashMap vs Hashtable vs ConcurrentHashMap

HashMap non-thread-safe; Hashtable synchronized; ConcurrentHashMap optimized thread safety.

30. HashSet vs TreeSet

HashSet unordered; TreeSet sorted.

Important Follow-Up Questions

HashMap Internal Working

Uses buckets, hashCode(), and equals() for storage and retrieval.

hashCode() and equals()

Ensure correct object comparison and hashing behavior.

Fail-Fast vs Fail-Safe

Fail-fast throws exception on modification; fail-safe works on a copy.

Comparable vs Comparator

Comparable defines natural ordering; Comparator custom ordering.

Generics

Provide type safety and remove explicit casting.

Collection Hierarchy

Collection -> List, Set, Queue; Map is separate.

Synchronization

Controls access to shared resources among threads.

Thread vs Runnable

Thread is a class; Runnable is an interface.

volatile

Ensures variable visibility across threads.

start() vs run()

start() creates new thread; run() executes normally.

Lambda Expressions

Compact syntax for functional programming.

Functional Interfaces

Interface with one abstract method.

Stream API

Processes collections declaratively.

Optional

Avoids NullPointerException.

Java 8 Features

Lambda, Streams, Optional, Method References, Date-Time API.

Reflection API

Examines/modifies classes at runtime.

Singleton Pattern

Ensures only one instance of a class exists.